

$$ax + by + c = 12$$

$$\left(\begin{array}{ccc|c} 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 2 \\ 1 & -1 & 1 & 0 \\ -1 & 1 & 1 & 1 \end{array} \right)$$

$\underbrace{\hspace{10em}}_A \quad \underbrace{\hspace{1em}}_{\vec{b}}$

$$A^* A \hat{x} = A^* b$$

$$\begin{pmatrix} 4 & -1 & 2 \\ -1 & 4 & 2 \\ 2 & 2 & 6 \end{pmatrix} \hat{x} = \begin{pmatrix} 0 & 1 & 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & 1 & -1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 2 \\ 0 \\ 1 \end{pmatrix}$$

$$\hat{x} = \begin{pmatrix} 4 & -1 & 2 \\ -1 & 4 & 2 \\ 2 & 2 & 6 \end{pmatrix}^{-1} \cdot \begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix}$$

$$\left(\begin{array}{ccc|ccc} 4 & -1 & 2 & 1 & 0 & 0 \\ -1 & 4 & 2 & 0 & 1 & 0 \\ 2 & 2 & 6 & 0 & 0 & 1 \end{array} \right) \sim \left(\begin{array}{ccc|ccc} 4 & -1 & 2 & 1 & 0 & 0 \\ 0 & 15 & 10 & 1 & 4 & 0 \\ 0 & 5 & 10 & -1 & 0 & 2 \end{array} \right) \sim$$

$$\begin{aligned}
 &\sim \left(\begin{array}{ccc|ccc} 4 & -1 & 2 & 1 & 0 & 0 \\ 0 & 3 & 2 & \frac{1}{5} & \frac{4}{5} & 0 \\ 0 & 1 & 2 & -\frac{1}{5} & 0 & \frac{2}{5} \end{array} \right) \sim \left(\begin{array}{ccc|ccc} 4 & -1 & 2 & 1 & 0 & 0 \\ 0 & 3 & 2 & \frac{1}{5} & \frac{4}{5} & 0 \\ 0 & 0 & 4 & -\frac{4}{5} & -\frac{4}{5} & \frac{6}{5} \end{array} \right) \sim \left(\begin{array}{ccc|ccc} 4 & -1 & 0 & \frac{3}{5} & \frac{2}{5} & -\frac{1}{5} \\ 0 & 3 & 0 & \frac{3}{5} & \frac{4}{5} & -\frac{1}{5} \\ 0 & 0 & 1 & -\frac{1}{5} & -\frac{1}{5} & \frac{3}{10} \end{array} \right) \\
 &\sim \left(\begin{array}{ccc|ccc} 4 & 0 & 0 & \frac{3}{5} & \frac{2}{5} & -\frac{1}{5} \\ 0 & 1 & 0 & \frac{1}{5} & \frac{2}{5} & -\frac{1}{5} \\ 0 & 0 & 1 & -\frac{1}{5} & -\frac{1}{5} & \frac{3}{10} \end{array} \right) \sim \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{3}{5} & \frac{2}{5} & -\frac{1}{5} \\ 0 & 1 & 0 & \frac{1}{5} & \frac{2}{5} & -\frac{1}{5} \\ 0 & 0 & 1 & -\frac{1}{5} & -\frac{1}{5} & \frac{3}{10} \end{array} \right)
 \end{aligned}$$

$$\hat{X} = \frac{1}{10} \begin{pmatrix} 4 & 2 & -2 \\ 4 & 4 & -2 \\ -2 & -2 & 3 \end{pmatrix} \begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix}$$

$$\hat{X} = \frac{1}{10} \begin{pmatrix} 4 \\ 12 \\ 6 \end{pmatrix} = \begin{pmatrix} \frac{2}{5} \\ \frac{6}{5} \\ \frac{3}{5} \end{pmatrix}$$

$$\underline{\underline{\mu = \frac{2}{5}x + \frac{6}{5}y + \frac{3}{5}}}$$